

Lee Yao

DATA ENGINEER · ANALYTICS ENGINEERING · AI SOLUTIONS

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SUMMARY

Data Engineer with experience building cloud-native data platforms, analytics pipelines, and AI-enabled data solutions. Skilled in designing end-to-end ELT workflows using **Python, Airflow, dbt, Snowflake, AWS, Spark, and Kafka**. Experienced in dimensional modeling, data quality frameworks, and integrating LLM-powered analytics into modern data platforms. Designed and implemented data platforms for healthcare, financial, e-commerce, and transportation domains using real-world datasets and production-grade architectures.

SKILLS

DATA ENGINEERING	ELT / ETL Pipelines(Python, SQL) · Orchestration (Airflow, Prefect) · Transformation (dbt) · Spark · Kafka · Snowflake · Cloud Ingestion (AWS S3) · Docker
DATA MODELING	Kimball Methodology · Star Schema · Fact & Dimension Modeling · Surrogate Keys· Data Marts
DATABASES	PostgreSQL · Snowflake · Neo4j (Graph) · DuckDB
PROGRAMMING	SQL · Python (Pandas, Matplotlib)
PLATFORM & CLOUD	Website Deployment · Cloud Hosting · API Integration · Workflow Automation
BI & ANALYTICS	Power BI · Qlik Sense
TOOLING	Docker · Git / GitHub · Power Automate · Power Apps · Excel
AI & ANALYTICS	Snowflake Cortex · OpenAI API · LLM-powered Analytics

EXPERIENCE

Solvía One LLC — Co-Founder & Data Platform Engineer

Full-time

May 2025 – Present

- Built data infrastructure and operational automation systems supporting a growing e-commerce business.
- Built cloud-based data workflows and reporting processes supporting product management, customer analytics, and business operations.
- Integrated LLM-powered content generation and workflow automation using OpenAI APIs to improve operational efficiency.
- Served as the primary technical contributor, designing and maintaining the company's data infrastructure, automation workflows, and cloud-based operational systems.

Freelance / Independent — Data Engineering & Analytics

Queens, NY

Jun 2024 – Present

- Built **end-to-end data engineering pipelines** across healthcare, e-commerce, and transportation domains, covering cloud ingestion (AWS S3), warehouse modeling (Snowflake, PostgreSQL), transformation (dbt), and orchestration (Airflow, Prefect).
- Implemented Kimball dimensional models and automated dbt data-quality testing across raw, staging, core, and mart layers, enabling reliable analytics-ready datasets and parameterized pipeline execution.
- Applied **Snowflake Cortex LLM** for multilingual sentiment analysis on 100K+ records and **Neo4j** graph modeling for fraud clusters and anomalous seller behavior.

SDG Group USA — Data Analyst

Bedminster, NJ

Dec 2022 – May 2024

- Developed and maintained **Qlik Sense and Power BI** dashboards for client reporting, operational monitoring, and decision-making.
- Investigated and resolved upstream data-quality and consistency issues, improving the reliability of Snowflake-powered reporting datasets feeding Power BI and Qlik Sense dashboards.
- Supported a BI improvement initiative for an airport-retail client — prepared reporting-ready datasets in Snowflake and resolved data-consistency issues feeding Power BI.
- Led a reporting migration for a pharmaceutical client: rebuilt Qlik Cloud reports in Power BI and automated daily distribution to stakeholders via Power Automate.

AMN Healthcare — Medical Interpreter (Part-Time)

Queens, NY

Sep 2025 – Present

- Provide real-time interpretation services in medical and business settings.

ADDITIONAL EXPERIENCE

Independent Insurance Broker — Client relationships, business development, and regulatory compliance

2016 – 2022

AI & DATA ENGINEERING PROJECTS

CMS Medicare Data Pipeline

April 2026 - May 2026

Kafka · Spark · Airflow · AWS S3 · Snowflake

- Built a healthcare data pipeline processing CMS Medicare datasets using **Kafka** for ingestion, **Spark** for distributed transformation, and **Airflow** for orchestration.
- Designed ingestion and transformation workflows integrating **Kafka**, **Spark**, **Snowflake**, and **Airflow** to automate healthcare data processing and scheduling.
- Implemented data quality validation and dimensional modeling to prepare analytics-ready datasets for downstream reporting.

GitHub ↗

Olist Supply Chain & AI Reviews Intelligence

Mar 2026 – Apr 2026

Python · Prefect · Docker · AWS S3 · Snowflake · Cortex LLM · dbt · Neo4j · Power BI

- Built an automated ELT pipeline (Python, Prefect, Docker) ingesting **100K+** e-commerce reviews and transactions from AWS S3 into Snowflake.
- Engineered multilingual (Portuguese) sentiment analysis in Snowflake using **Cortex LLM** with SQL-based fallback logic for consistent large-scale processing.
- Applied **Neo4j graph modeling** to surface suspicious review patterns and detect fraud clusters among sellers.
- Designed a Kimball star schema in dbt with automated data-quality tests; built an executive Power BI dashboard showing the bottom 3% of sellers drove **40%+** of negative sentiment.
- Linked logistics SLA breaches across 5 regions to a 3.5-point satisfaction drop, enabling targeted interventions.

GitHub ↗

US DOT Flights Orchestrated Data Pipeline

Apr 2026 – May 2026

Apache Airflow · Docker Compose · Python · AWS S3 · Snowflake · dbt

- Engineered an end-to-end **monthly pipeline** (Dockerized Airflow, Python, AWS S3, Snowflake) ingesting and transforming US DOT (BTS) flight data from raw download to analytics-ready marts.
- Orchestrated parameterized DAGs — download, GZIP, S3 upload, Snowflake COPY INTO, dbt run/test — with **idempotency checks** at every stage for safe monthly backfills.
- Established a Snowflake external stage via S3 Storage Integration (IAM trust chain) for serverless loading into raw / staging / core / mart layers.
- Modeled dbt fact/dim tables (fact_flights, dim_carrier, dim_airport, dim_date with dynamic date spine) using MD5 surrogate keys for join-efficient dimensional modeling.
- Containerized the full Airflow stack (webserver, scheduler, metadata DB) with Docker Compose for a reproducible, isolated environment.

GitHub ↗

NYC Taxi Analytics Engineering Pipeline

Dec 2025 – Mar 2026

PostgreSQL · dbt · Docker · Power BI

- Built an end-to-end ELT pipeline transforming raw NYC Taxi trip data into analytics-ready reporting tables.
- Designed layered models across raw / staging / core / mart schemas; implemented data-quality validation for anomalies like unrealistic trip speeds and invalid timestamps.
- Created daily aggregated marts (trip volume, revenue, duration) and Power BI dashboards for demand, revenue, and airport-traffic analysis.
- Built a Dockerized PostgreSQL + dbt development environment for reproducible analytics engineering workflows.

GitHub ↗

EDUCATION

Stony Brook University — B.S., Electrical Engineering

Stony Brook, NY · May 2008

CERTIFICATION

Microsoft Certified: Power BI Data Analyst Associate

Expires Jul 2026